

The Online Algorithmic Complexity Calculator

v3.0

To know how to calculate your personal 'cognitive randomness' ability (as shown in our widely covered article) read [this](#).
The data produced by more than 3400 people trying to generate random data can be found [here](#) (make sure to cite properly as explained [here](#)).
For the new functionality on network analysis read:
[An Algorithmic Information Calculus for Causal Discovery and Reprogramming Systems](#)
and also, very important, the [Numerical Limitations](#) subsection in the [How It Works](#) subpage.

For any string

For binary arrays/adjacency matrices

For short strings

Network perturbation

Block Decomposition Method for Strings

Enter a string

0101010101010101010101010101010101

Block size

2

12

Block overlap

0

11

Alphabet size

☒ 2

☐ 4

☐ 5

☐ 6

☐ 9

☐ 256 (utf-8)

Evaluate

Result of Evaluation

$$BDM = \sum_{i=1}^n K(\text{block}_i) + \log_2(|\text{block}_i|)$$

Strings that don't appear in the $D(\# \text{ of states}, \# \text{ of symbols})$ distribution have their K value estimated as $\text{Max}(K(\# \text{ of states}, \# \text{ of symbols})) + 1$

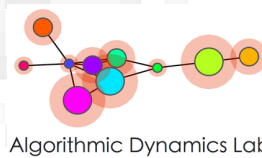
The [Online Algorithmic Complexity Calculator](#) (OACC) is a powerful tool that provides estimations of [algorithmic complexity](#) (a.k.a. [Kolmogorov-Chaitin complexity](#)) (or $K\$$) and [Algorithmic Probability](#) for short and long strings and for 2-dimensional arrays better than any other tool, and estimations of [Bennett's Logical Depth](#) (or $LD\$$) for strings, hitherto impossible to estimate with any other tool. These 3 measures constitute the most powerful and universal algorithmic measures of complexity.

To estimate the complexity of long strings and large adjacency matrices, the OACC uses a method called $\$BDM\$$ which is based upon [Algorithmic Probability](#). The estimation of complexity by $\$BDM\$$ is compatible with --but beyond the scope of-- lossless compression algorithms, which are so widely used to estimate $K\$$. [Implementations of lossless compression are, however, entirely based upon Shannon entropy \(\$\\$SS\\$\$ \)](#) (e.g. [LZ](#), [LZW](#), [DEFLATE](#), etc) and thus cannot capture any algorithmic content beyond simple statistical patterns (repetitions). In contrast to lossless compression, $\$BDM\$$ not only considers statistical regularities but is also sensitive to segments of an algorithmic nature (such as sequences like $\$12345...\$$ or $\$246810...\$$), which Shannon entropy and lossless compression algorithms would only be able to characterize as having maximum randomness and the highest degree of incompressibility. Moreover, unlike Kolmogorov-Chaitin complexity (thanks to the [Invariance Theorem](#)), both Entropy and Entropy-based compression algorithms **are not invariant** to language choice and are therefore not robust enough to measure complexity or randomness ([technical arguments and an example are be found here](#)).

How to cite the OACC if you use it

For exhaustive or heavy calculations please do download or use alternatives such as our [R implementation](#). Otherwise the bandwidth will reach its limit and other people will not longer be able to use the calculator until the next monthly period.

The OACC is a project developed by the following labs:



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